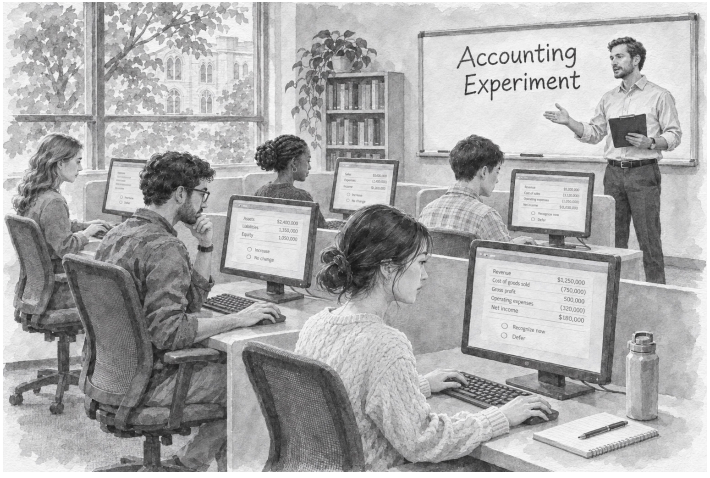
Recovering Accounting Judgment with AI: Textual and Non-Textual Experiments

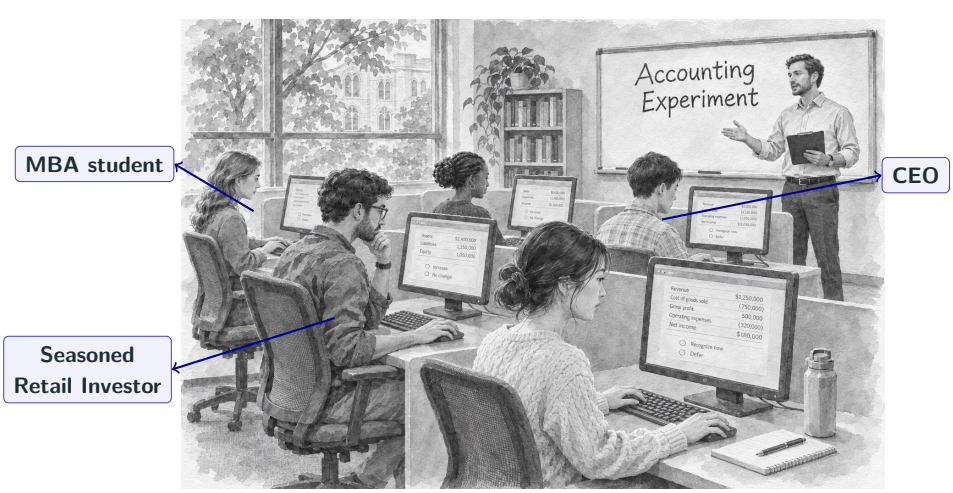
Hunter Ng

November 2026

AI as human participants



AI as human participants



AI as human participants

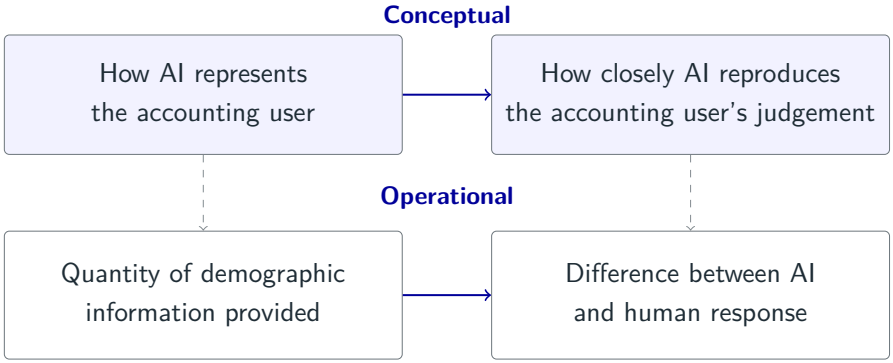


AI as human participants



- Using Published Experiments
- Giving AI the tests
- Can we reproduce the human response?

Libby Box



Adapted from Libby, Bloomfield, and Nelson (2002).

Motivation (I) - Background



- Dystopian future
- "Voter of the Year"
- No need for Elections

Motivation (I) - Background



Dystopian future	Present Reality
"Voter of the Year"	One AI Represents Many Profiles
No need for Elections	No need for Real Human Responses

Motivation (II) - Silicon Sampling

Silicon sampling

Give AI a participant profile and ask it the same questions you would ask a person. Use several profiles to create a sample.

Motivation (II) - Silicon Sampling

Silicon sampling

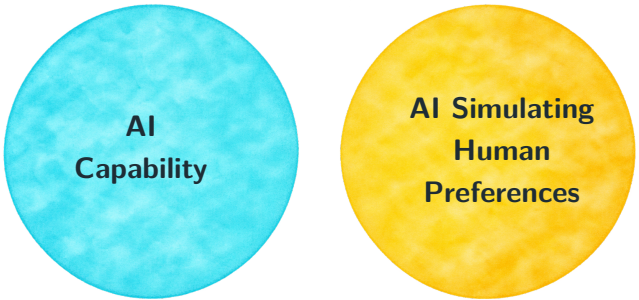
Give AI a participant profile and ask it the same questions you would ask a person. Use several profiles to create a sample.

- Synthetic-user startup Simile raised **$300 million** across two funding rounds in 2026.
  TechCrunch (July 2026)
- Kenvue used synthetic consumer feedback to help develop Aveeno's **Sensitive Skin Scents** line.
  Digiday (October 2026)

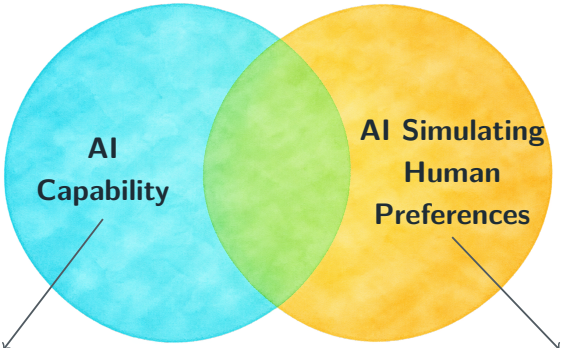


Aveeno oat face cleanser (illustration)

Motivation (III) - Accounting Judgement



Motivation (III) - Accounting Judgement



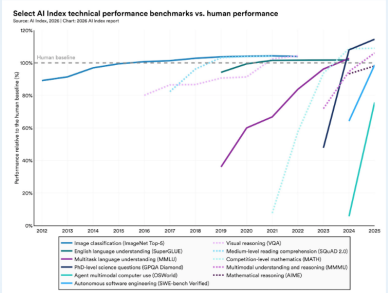
AI Capability Is Increasing

Top Takeaways

1. AI capability is not plateauing. It is accelerating and reaching more people than ever.

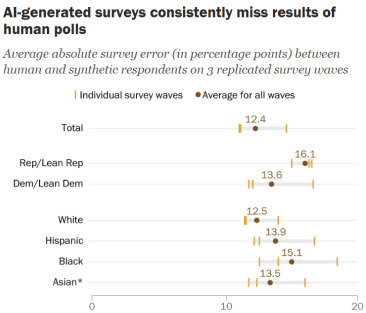
Industry produced over 90% of notable frontier models in 2025, and several of those models now meet or exceed human baselines on PhD-level science questions, multimodal reasoning, and competition mathematics. On a key coding benchmark—SWE-bench Verified—performance rose from 60% to near 100% in a single year. Organizational adoption reached 88%, and 4 in 5 university students now use generative AI.

See: Chapter 2: Technical Performance



+30 percentage points in one year on Humanity's Last Exam. Stanford AI Index (April 2026)

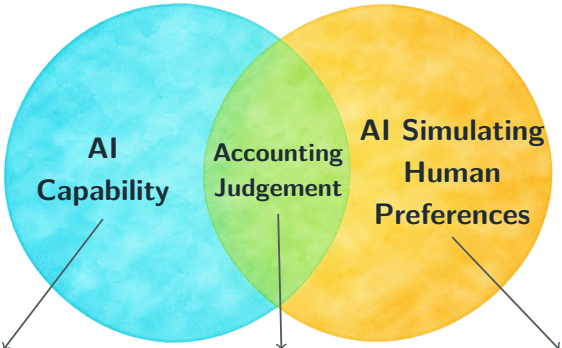
AI Survey Accuracy Can Improve



- **Richer profiles improved AI survey accuracy.** Adding political attitudes and values reduced average error from 16.0 to 13.7 percentage points on the same survey.

Pew Research Center, September 2026

Motivation (III) - Accounting Judgement



Setting

- **Published Standard-Setting Experiments**
  - **Sources:** Koonce et al. (2024), supplemented by published articles, appendices, and experimental materials.
  - **Coverage: 395 distinct findings from 64 papers** published from **1995 through 2025**.
  - **ASC areas:** Presentation (200s), Assets (300s), Liabilities (400s), Revenue (600s), Expenses (700s), Broad Transactions (800s), and Industry (900s).
- **AI Model**
  - **Primary model: GPT-6 Luna**, with temperature set to zero and reasoning disabled.
  - **Inputs:** Reconstructed cases, treatment conditions, and questions.
  - **Robustness tests:** Check whether documented LLM behaviours affect the results.

Koonce et al. (2024): Sample Papers by Standard

Conceptual framework

2006 Frederickson et al.
2010 Hodge et al.
2010 Koonce et al.
2011 Koonce et al.
2012 Kadous et al.
2014 Bonner et al.
2015 Anderson et al.
2015 Hewitt et al.
2016 Majors
2020 Doxey et al.
2021 Young
2022 Barradale et al.
2022 Liu et al.

ASC 700 Expenses

1995 Cloyd
2001 Luft & Shields
2004 Hodge et al.
2006 Frederickson et al.
2009 Clor-Proell
2017 Emett & Nelson

ASC 200 Presentation

1998 Hirst & Hopkins
2000 Maines & McDaniel
2001 Hronsky & Houghton
2004 Hirst et al.
2006 Bloomfield et al.
2008 Hodder et al.
2010 Elliott et al.
2010 Hodge et al.
2014 Clor-Proell et al.
2015 Elliott et al.
2015 Hewitt et al.
2015 Lachmann et al.
2017 Emett & Nelson
2019 Koonce et al.

ASC 900 Industry

2001 Dietrich et al.
2003 Hirst et al.
2010 Koonce et al.
2021 Eilifsen et al.

ASC 300 Assets

1996 Cloyd et al.
1998 Hirst & Hopkins
2000 Hopkins et al.
2000 Maines & McDaniel
2001 Luft & Shields
2006 Bloomfield et al.
2008 Jackson
2009 Clor-Proell
2010 Elliott et al.
2010 Jackson et al.
2015 Elliott et al.
2015 Rennekamp et al.
2017 Asay et al.
2018 Tan & Trotman
2020 Kunz & Staehle

ASC 600 Revenue

2007 Clor-Proell & Nelson
2012 Bailey & Sawers
2021 Young

ASC 400 Liabilities

1998 Kennedy et al.
2006 Hodge et al.
2014 Clor-Proell & Maines
2015 Fanning et al.
2016 Clor-Proell et al.
2018 Kelton & Montague
2020 Du et al.

ASC 800 Transactions

1995 Cloyd
2000 Hopkins et al.
2004 Hirst et al.
2005 Koonce et al.
2007 Nelson & Tayler
2010 Jackson et al.
2010 Jamal & Tan
2011 Agoglia et al.
2011 Gaynor et al.
2011 Koonce et al.
2012 Bailey & Sawers
2012 Hales et al.

ASC 800 (continued)

2012 Kadous et al.
2013 Chen et al.
2013 Cohen et al.
2014 Bonner et al.
2014 Clor-Proell et al.
2014 Griffin
2015 Anderson et al.
2015 Lachmann et al.
2015 Nelson & Rupar
2016 Majors
2019 He et al.
2019 Koonce et al.
2020 Doxey et al.
2020 Tan & Mayorga
2021 Chen et al.
2021 Eilifsen et al.
2022 Barradale et al.
2022 Liu et al.

Example: Cloyd (1995)

How do tax preparers advise clients?



- 72 experienced tax professionals.
- The client's appetite for tax risk was changed.
- Preparers backed aggressive tax positions more strongly when treatments matched or the client was aggressive.

AI Prompt

Participant role

Case instructions

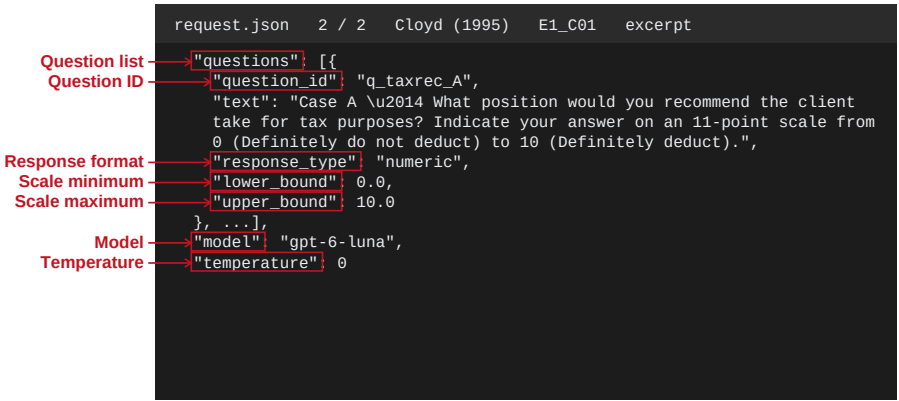
Client risk

Case A excerpt

```
request.json   1 / 2   Cloyd (1995)   E1_C01   excerpt

{
  "participant_role": "Intended role: an experienced tax preparer with at
    least tax-senior-level experience; the source describes recruiting
    tax seniors and managers.",
  "material": "Work independently and treat the two hypothetical cases as
    independent. ...
    Both clients are generally aggressive, risk-taking taxpayers. ...
    Case A \u2014 In March 1988, PARENTCO formed subsidiary SUBCO to
    construct and operate a new widget manufacturing facility. ...",
  ...
}
```

AI Prompt


request.json 2 / 2 Cloyd (1995) E1_C01 excerpt
"questions": [{
"question_id": "q_taxrec_A",
"text": "Case A \u2014 What position would you recommend the client take for tax purposes? Indicate your answer on an 11-point scale from 0 (Definitely do not deduct) to 10 (Definitely deduct).",
"response_type": "numeric",
"lower_bound": 0.0,
"upper_bound": 10.0
}, ...],
"model": "gpt-6-luna",
"temperature": 0
Question list
Question ID
Response format
Scale minimum
Scale maximum
Model
Temperature

AI Prompt

Question ID
Model explanation
Answer on scale

response.json Cloyd (1995) E1_C01 selected fields

```
{ "answers": [{
    "question_id": "q_taxrec_A",
    "answer_text": "Recommend deducting the management fee in 1989; the
    ongoing efforts to arrange future sales and the profit-making purpose
    support an active trade or business, though the delayed facility
    opening creates uncertainty.",
    "numeric_answer": 7
  },
{"question_id": "q_audit_A", "numeric_answer": 10},
{"question_id": "q_defense_A", "numeric_answer": 60},
{"question_id": "q_taxrec_B", "numeric_answer": 6},
{"question_id": "q_audit_B", "numeric_answer": 10},
{"question_id": "q_defense_B", "numeric_answer": 55}
] }
```

My Dataset

Manually encoded from the sample in Koonce et al. (2024)



Sample Selection

	Findings	Papers
Published catalogue	**395**	64
Remove findings with unavailable materials or unsupported designs	(132)	—
Remove unusable scores	(27)	—
Eligible baseline direction tests	**236**	46
Require verified human means or proportions by condition (net)	(152)	—
Findings with verified human means for every condition	**84**	23
Require scorable AI answers for the same conditions and documented scales	(36)	—
Judgement levels, knowledge vs. neutral	**48**	**16**

From Papers to Findings and Questions

64 experimental papers



Each icon represents one paper.

Example from Cloyd (1995)

Published finding **395 in the catalogue**

Preparers backed the aggressive tax position more strongly for a risk-tolerant client.

Factual question

What management fee did SUBCO pay MANCO in 1989? **$1 million.**

One paper can contribute several findings and several factual questions.

From Papers to Findings and Questions

What papers report A regression can show which condition scored higher. To compare levels, we also need the human mean or response proportion in each condition.

132 excluded before direction scoring If Cloyd reported an effect of tax and financial reporting conformity but omitted the case details that defined the two conditions, we could not recreate the comparison.

Direction versus magnitude Suppose Cloyd reported only that conformity raised the estimated chance of defending the deduction. We would know the direction. To score levels, we need the average estimate in each condition.

What is a scorable AI rating? A number we can compare directly with the human average. In Cloyd's tax case, GPT-6 gave a **7 out of 10** for recommending a deduction. If the human average were **6 out of 10**, the AI's error would be **1 point**. "I would recommend the deduction" gives us no number to compare. We need a usable rating for every condition.

Survey of Consumer Finances



- A Federal Reserve survey of U.S. families every three years.
- 1983 wave. Current series since 1989. About 4,400–6,300 families per recent wave.
- Tracks income, assets, debts, pensions, and risk.
- **Four risk choices**
  (1) No risk (2) Average risk
  (3) Above-average risk (4) Substantial risk
- I weight the four profiles using investor shares from the preceding survey.

Federal Reserve Board (SCF) Questionnaire

Finance studies using the SCF

Campbell (2006, *JF*), Becker and Shabani (2010, *RFS*), Gálvez and Paz-Pardo (2026, *JFE*)

(I)

Can Profiles Improve Recovery of Accounting Judgement?



Participant Profiles and Survey Weights

Change in human-direction recovery (percentage points)

	(1) Equal risk mix vs neutral	(2) Historical mix vs neutral	(3) Historical vs equal mix
Estimate (pp)	11.83**	13.44**	**1.61**
Paper-clustered SE	(5.48)	(5.43)	(1.69)
Findings	186	186	186
Controls	Finding FE	Finding FE	Finding FE

$100(R_{i,A} - R_{i,B}) = \beta_{A-B} + \varepsilon_i \quad R_{i,a} = 1$ for a match to the human direction, 0 otherwise.

Combining four risk profiles raises recovery from **42% to 54%**. Weights from the closest preceding SCF wave bring recovery to **55%**, an estimated **1.61-point** increment.

What We Actually Sent to the Model

Cloyd (1995) Tax-preparer judgement

Prompt component	Excerpt or instruction
Original role	Experienced tax preparer with at least tax-senior-level experience.
Original case	Work independently and treat the two hypothetical cases as independent. In this condition, both clients are aggressive, risk-taking taxpayers.
Original question	"What position would you recommend the client take for tax purposes?" Rate it from 0 (definitely do not deduct) to 10 (definitely deduct).
Added knowledge	"For this task, you are assigned strong knowledge of tax accounting."

The neutral prompt keeps the original tax-preparer role, case, and questions. The knowledge prompt adds the last line.

(II)

Does AI Recover Accounting Judgement Better in Textual or Visual Experiments?



Textual and Visual Recovery Across Models

How is the treatment presented?

Treatment	What changes	Findings
Textual	Wording, labels, instructions, or amounts in a fixed layout.	163
Visual	A table, graph, or financial-statement layout.	53

Textual example. Cloyd (1995) varied how the written case described the client's willingness to take tax risks.

Visual example. Hirst et al. (2004) showed the same bank under full or piecemeal fair-value reporting.

Textual and Visual Recovery Across Models

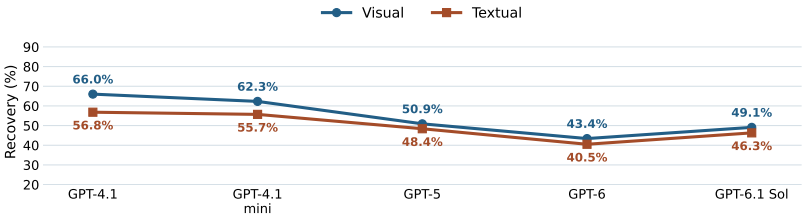
Does the treatment change the underlying information?

Treatment	What changes	Findings
Information change	New facts or different economics.	17
Presentation only	Same information, different display.	29

Information change. Hirst et al. (2004) compared a bank that hedged its interest-rate exposure with one that stayed exposed.

Presentation only. Bonner et al. (2014) showed either one $70,000 gain or a $1 million gain and a $930,000 loss. The net gain was the same.

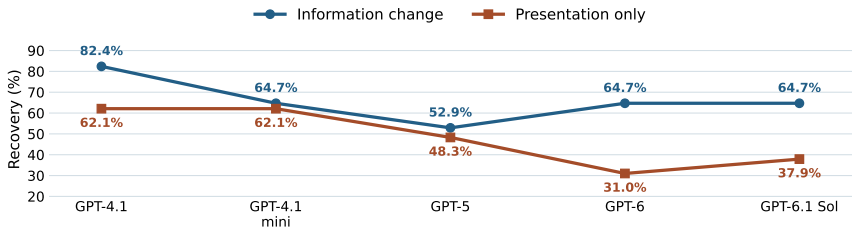
Textual and Visual Recovery Across Models



	GPT-4.1	GPT-4.1 mini	GPT-5	GPT-6	GPT-6.1 Sol
Visual minus textual (pp)	**9.2**	6.6	2.5	**2.9**	2.8
Paper-clustered SE (pp)	(11.34)	(11.25)	(9.02)	(11.64)	(12.04)
Textual findings (papers)	162 (36)	158 (36)	159 (35)	163 (36)	162 (36)
Visual findings (papers)	53 (14)	53 (14)	53 (14)	53 (14)	53 (14)

The estimated gap narrows from 9.2 points for GPT-4.1 to 2.9 for GPT-6.

Textual and Visual Recovery Across Models



	GPT-4.1	GPT-4.1 mini	GPT-5	GPT-6	GPT-6.1 Sol
Information minus presentation (pp)	20.28	2.64	4.67	33.67*	26.77
Paper-clustered SE (pp)	(15.11)	(20.44)	(15.37)	(18.99)	(16.86)

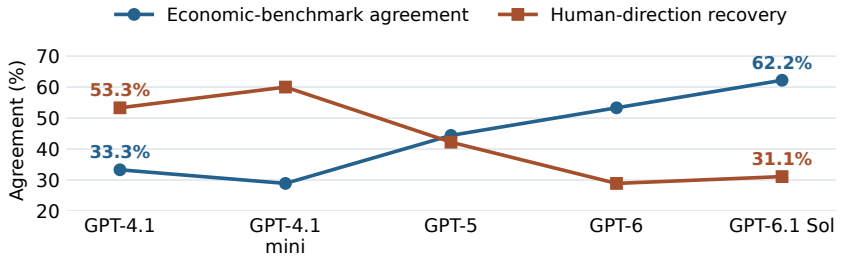
All models use 17 information-change findings (6 papers) and 29 presentation-only findings (11 papers). The groups span 14 distinct papers. $^{*}p < 0.10$ (unadjusted).

(III)

Can Economically Appropriate Answers Differ From Human Judgments?



Economic Agreement and Human Recovery



Same 45 findings from 13 papers for all five model configurations.

Economic benchmark. The direction implied by the task's economics.

Example. In Hirst et al. (2004), the same bank had the same interest-rate exposure under full and piecemeal fair-value reporting. The benchmark predicts the same investment-risk rating. Analysts rated it riskier under full fair value.

Example: Same Bank, Two Reporting Formats

Hirst et al. (2004). The same bank had the same interest-rate exposure. Only the reporting format changed.

Economic benchmark	The authors' economic prediction was **no difference** in investment risk.
Human ratings	Mean risk was **10.10** under full fair value and **8.11** under piecemeal fair value, on a 1–15 scale.

(IV)

Robustness: Could the Model Be Recalling Published Results?



Robustness: Look-Ahead Bias

Closed-book recall	Result
Exact published numbers matched	**0 / 1,894**
Findings	236

(V)

Robustness: Do the Profiles Work on a New Study?



Le (2026) Accounting Rules and Career Choice

Le tests whether strict accounting rules deter students from accounting careers. Table 9 covers 811 Columbia and Prolific undergraduates randomly assigned to read about GAAP as strict rules, as a guide to judgment, or to read no description. They rated each career deterrent on a 1–5 scale.

Human estimates from Table 9

Deterrent	Judgment-based vs. neutral	Restrictive vs. neutral	Restrictive vs. judgment-based
Limited creativity or critical thinking	−0.041	**+0.194**	**+0.235**
Work is too difficult	−0.154	−0.172	−0.018
Pay is insufficient	−0.008	−0.008	0.000
Too many requirements	−0.116	−0.142	−0.026

Robustness: Out-of-Sample Accounting Judgments

Le (2026)	Neutral prompt	Weighted demographics
Recovered human directions / 11	2/11	**4/11**

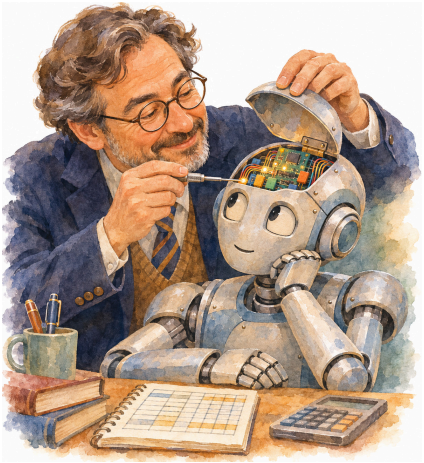
Le (2026) tests whether describing GAAP as strict rules makes accounting less attractive to undergraduates who value creative work.

(VI)

What Does AI Know About Accounting Judgements?



How the Probe Reads Internal States



- **Prior work.** Jahanparast et al. use a linear probe to predict group survey responses from internal states.
- **Read the model.** Save Qwen3-8B's internal states just before it answers the accounting question, along with its written answer.
- **Fit the probe.** Use other papers' human ratings to train a linear predictor with and without internal states.
- **Test on a held-out paper.** Keep Qwen3-8B's weights fixed.

Jahanparast, Hong, and Chang (ICLR 2026)

Probing Internal Representations

Prediction method	Error (% of response scale)
Written model answers	20.0%
Calibrated written answers	10.4%
Calibrated answers + internal states	**9.8%**
Findings with documented scale bounds	27

We adapt the linear probing approach of Jahanparast et al. (2026). We keep Qwen3-8B fixed, fit a predictor of human ratings using its answers and internal states, and test it on papers held out of training.

Improvement. Adding internal states lowers average error from 10.4% to 9.8% (0.6 percentage points).

Next Steps

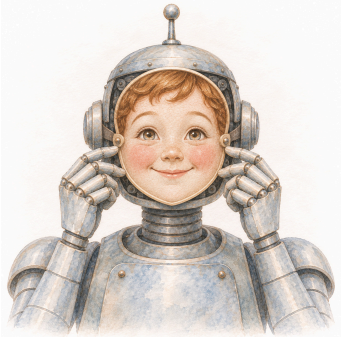
- Build an online interface with profile weights ready to use.
- Let researchers switch between equal weights and survey weights by year and test their own accounting experiments.
- Extend the tests to more new accounting studies.

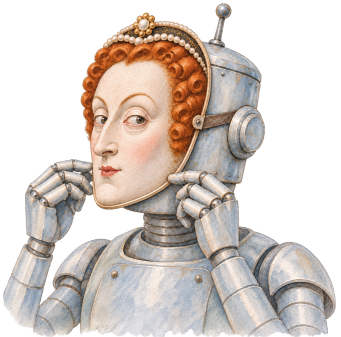
Conclusion

- **Main result.** Four risk profiles raise human-direction recovery from **42% to 54%**. Survey weights from the study period bring it to **55%**.
- **Presentation.** Recovery shows no clear difference between textual and visual experiments. Within visual experiments, GPT-6 recovers **65%** when information changes and **31%** when only presentation changes.
- **Facts and judgement.** AI can get the accounting facts right and still respond differently from people.
- **Economic agreement and human recovery.** Neither measure shows a steady trend across the five models.
- **Internal states.** Adding internal states reduced held-out rating error from **10.4% to 9.8%** of the response scale across 27 findings.

Thank you







Useful information requires accounting judgement

Framework characteristic	Requirement
Relevance	Predictive or confirmatory value for decisions.
Faithful representation	Complete, neutral, and free from error.
Enhancing qualities	Comparability, verifiability, timeliness, understandability.
Cost constraint	Reporting benefits justify the costs.

Estimates incorporate future expectations and estimation error (Barth, 2006; Dechow and Dichev, 2002). Evaluating complex estimates requires integrating evidence (Griffith et al., 2015).

The source catalogue and the analytical samples

Analytical population	Count	Papers
Source catalogue: 401 accepted records	395	64
Source-compatible nonzero directions	208	47
Controlled procedure patterns	45	28
Unprofiled model-comparison panel	236	46
Findings with complete human mean benchmarks	84	23

The inventory contains 65 studies. Paired tests use the same available findings across the compared profiles.

Participant profiles describe the decision maker

Profile family	Profiles	Description
Reference prompts	2	Original prompt; fresh neutral prompt
Assigned knowledge	1	Accounting expertise
Uncertainty and evidence	8	Cautious/open attitudes; higher/ordinary thresholds; with or without knowledge
Other characteristics	8	Horizons, familiarity, attention, objectives
Financial risk	4	SCF investment-risk categories

Primary profile model: GPT-6 Luna; temperature 0; reasoning disabled.

Four risk profiles define the audience mixture

Substantial risk; above-average risk; average risk; no financial risk.

$$\bar{y}_g(t) = \sum_{j=1}^{4} w_{gj} y_j(t), \qquad \sum_{j=1}^{4} w_{gj} = 1.$$

Audience mixture	Shares applied to the same four responses
Equal mixture	25% in each risk category
Dated audience	Closest preceding Survey of Consumer Finances wave
Fixed audience	2022 Survey of Consumer Finances distribution

Directions, judgement levels, and economic decisions

Outcome	Evaluation criterion
Human recovery	Same sign as the source-compatible nonzero human contrast
Controlled pattern agreement	Source-defined pattern under the adapted task procedure
Judgement-level agreement	Distance from published human condition means
Economic agreement	Source-supported accounting rule or economic decision

Separate factual checks evaluate amounts, classifications, and relationships.